

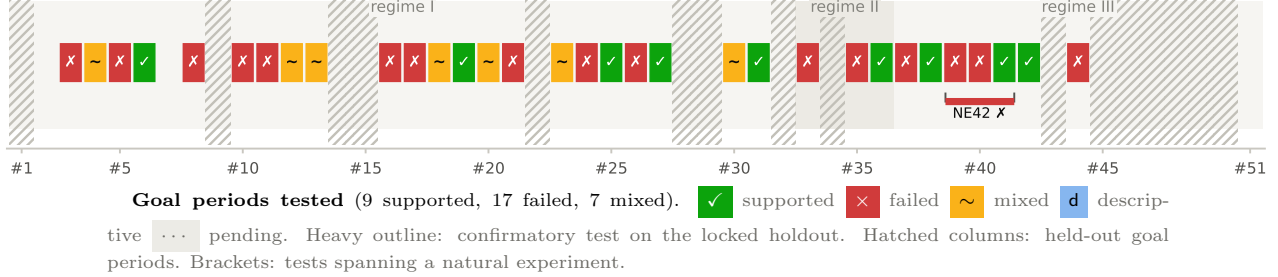
H32: A net information current identifies de facto leaders

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 1b)

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

Bottom line. Exposure-gated content transfer exists in about half the periods and gives a reproducible ranking, but elected, installed and operator leaders all rank low.

level: hypothesis · pipeline checks 86% · updated 2026-10-04 (round 1b)



WHAT HAPPENED

H32 asks whether agent i 's recent message, seen by j , predicts the direction of j 's next message. The baseline is j 's own past, the fields and the rest of what j saw. Held-out-day cross-validation tests the gain against circular-shift nulls on 32 non-holdout periods (#3–#44). Transfer exists but is small: it is significant in 18/32 periods (15/32 against the stricter within-day null). It needs exposure: seen > unseen in 7/8 multi-room weeks. The outflow ranking replicates across halves of days (median $\rho = 0.38$). It does not find leaders: a single source stands out in 2/18 periods, and humans beat the median agent in 4/16. Round 1b uses context-ledger exposure, a second embedding model and style removal. Transfer replicates (17/32; gte 23/32). At matched short lag, read messages beat in-flight ones in 17/17 transfer periods. In-flight messages still carry about a third of the gain (convergence). All three leader tests fail: the elected leader over its real term (rank 5/10), #35's daily lead designers, and #44's installed leader (last of 6). The holdout is not run.

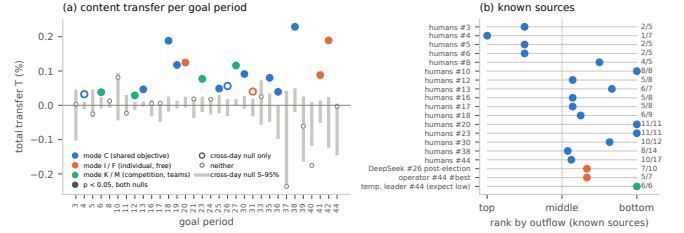
PHYSICAL MODEL

Models 11 and 02 give a message-level vector autoregression on whitened embeddings with isotropic couplings. The state $\mathbf{z}_m$ is agent j 's message vector after removal of a 5-direction goal, kickoff and day×room field. The source $\mathbf{s}_{ij}$ is the decayed sum ($\tau = 15$ min) of i 's messages that j had seen. $\mathbf{B}_{m,c}$ are the baseline columns: own past, day×room field, other seen senders, human and bot messages. G_{ij} is the held-out gain (Gaussian transfer entropy). The order parameter Φ is the noise-corrected concentration of outflow.

$$\mathbf{z}_m = \sum_c \beta_c \mathbf{B}_{m,c} + \kappa_{ij} \mathbf{s}_{ij}(t_m) + \xi_m, \quad \mathbf{s}_{ij} = \frac{\sum_{x \in \text{seen}} e^{-\Delta t_x / \tau} \mathbf{z}_x}{1 + \sum_{x \in \text{seen}} e^{-\Delta t_x / \tau}}$$

$$G_{ij} = 1 - \text{SSE}_{\text{full}} / \text{SSE}_{\text{base}} \text{ (held-out days),} \quad \text{TE}_{i \rightarrow j} \approx -\frac{1}{2} \ln(1 - G_{ij})$$

$$\text{Out}_i = \langle \Delta G_{ij} \rangle_j, \quad \text{Net}_i = \text{Out}_i - \text{In}_i, \quad \Phi = \frac{\text{Var}_i(\text{Out}) - \sigma_{\text{null}}^2}{(N-1) \overline{\text{Out}}^2}$$



Observables. (a) Total transfer T (mean pair gain minus the null median) per period. Gray bars: cross-day null 5–95%. Filled: significant against the cross-day and the within-day null. Ringed: cross-day only. (b) Rank of known sources by outflow. Humans (blue) are top sources only in the viewer-heavy first weeks. The elected (#26) leader, the installed (#44) leader and the #44 operator are not top sources.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not assume that the agent you put in charge drives the content. In this village, formal leaders, elected or fine-tuned, are not content sources. The installed leader is the most-influenced agent in its room, as H02 and H23 also find. Most weeks are distributed: content flows among several agents, often the strongest Claude model present. A monitor can compute total transfer T , which shows whether agents take content from each other, and Φ , which shows whether one source stands out. In one room, a fast responder to outside news looks the same as a leader. The top name changes with the embedding model (17/32 agree). Use T as a coupling monitor, not as an org chart.

H32-information-current-leaders · generated 2026-10-04 12:00 by infra/summaries/build_summaries.py · card hypotheses/H32-information-current-leaders/README.md · ratings: H32 agent (suggested); round 1b RE-D1 (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 transfer in $\geq 60\%$ of periods	18/32 (56%); 15/32 under both nulls	failed (narrowly)
P2 a source stands out in $\geq 50\%$	2/18 transfer periods	failed
P3 split-half $\rho > 0$ in $\geq 2/3$, median ≥ 0.3	23/26, median 0.38; 1b: 20/26, 0.27	supported (round 1)
P5 humans above the median agent in $\geq 70\%$	4/16 (only #3–#6); 1b: 6/29	failed
P6b #26 elected leader top after the vote	rank 7/10 ($z = 0.1$)	failed
P7a #44 installed leader not top	last of 6 agents	supported
P7b #44 operator top in #best	5th of 7 sources; 1b: 6th of 7	failed
P8 seen > unseen (multi-room)	7/8 weeks, sign $p = 0.035$; unseen sig. 1/8	supported
P9 NE42 merge switches transfer on	+0.0001%, sign $p = 0.39$	failed
1b: read > in-flight at same lag	17/17; in-flight sig. 13/17, $\approx 1/3$ of the gain	mostly copying
1b natives: #26 term, #35 leads, #44 window	rank 5/10; percentile 0.43; last of 6	failed (as predicted)
P10–P12 modes and rivals	$C > I/F$ $p = 0.10$; Φ by mode $p = 0.42$; $ \rho $ with timing 0.24	mixed

GOAL PERIODS IN DETAIL

G03 $\times$ 1b: failed (ledger exposure: T +0.007% p 0.341; gte p...
G04 $\sim$ 1b: mixed (ledger exposure: T +0.028% p 0.024, split...
G05 $\times$ 1b: failed (ledger exposure: T -0.026% p 0.927; gte p...
G06 $\checkmark$ 1b: supported (ledger exposure: T +0.037% p 0.024; gte...
G08 $\times$ 1b: failed (ledger exposure: T +0.012% p 0.220; gte p...
G10 $\times$ 1b: failed (ledger exposure: T +0.085% p 0.122, split...
G11 $\times$ 1b: failed (ledger exposure: T -0.024% p 0.951, split...
G12 $\sim$ 1b: mixed (ledger exposure: T +0.015% p 0.049, split...
G13 $\sim$ 1b: mixed (ledger exposure: T +0.042% p 0.024, split...
G16 $\times$ 1b: failed (ledger exposure: T -0.001% p 0.415, split...
+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

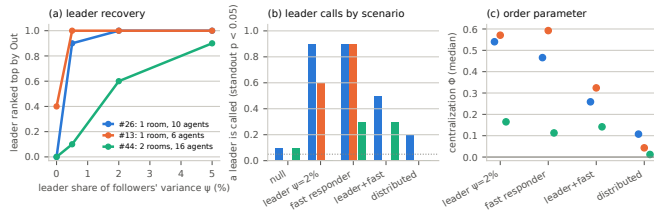
The effects are fractions of a percent of residual variance. With 32 periods and 13 predictions, some passes are chance. In one room, a fast responder and a leader look the same; only the multi-room exposure contrasts and the within-day null address this. The method measures sustained conversational influence, not one-shot agenda setting such as the #26 goal announcement. Outflow is a total, so low-volume sources (operators) rank low. Transfer depends on the number of field directions removed: 25, 18 and 7 significant periods for 1, 5 and 10. The top source agrees in only 13–27 of 32 periods across preprocessing variants, and in 17/32 across embedding models (1b). In regime III, low-volume targets make leave-one-day-out unstable (post-hoc A2). Part of the short-lag transfer is two agents that answer the same prior turn (1b).

ROUND 2

Run the frozen holdout test (#28, #22, #14). Add a per-message coupling, so the test judges operators per message. Add an event-study variant for one-shot agenda setting. Use the #51 multi-room segments and NE38 (an operator reassigns one agent) for exposure and field contrasts with more power.

No round-2 redirects yet.

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Second figure. Synthetic validation on real message skeletons. (a) The estimator ranks a planted leader top in single-room weeks at 0.5% of the followers' variance. The two-room week needs about 5%. (b) False calls are 0–10%. In one room, the estimator calls a fast responder to a hidden drive the leader 90% of the time. (c) Φ separates a star from a distributed swarm.